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VLM-Augmented Sovereign AI Desktop Assistant

CSE420 – Graduation Project 1 – Thesis

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Abstract

The rapid proliferation of cloud-based artificial intelligence assistants has introduced systemic risks regarding data privacy, latency, and operational costs. This thesis presents the comprehensive design, implementation, and rigorous evaluation of a **Sovereign AI Desktop Assistant** — a fully local, privacy-preserving system that addresses the critical research question: *Can an embedded vision-language model pipeline, running entirely on local hardware, match the intelligence of cloud-based counterparts while guaranteeing that no user data ever leaves the device?*

This project successfully demonstrates a novel two-stage architecture. In the first stage, an ESP32-S3-N16R8 microcontroller acts as the edge capture node, handling real-time image acquisition via an OV2640 camera and audio streaming through I2S, encoding this multimodal data efficiently within its constrained memory resources (specifically leveraging 8 MB PSRAM). The second stage delegates complex reasoning to a locally hosted Large Vision-Language Model (VLM), specifically `llama3.2-vision` served via the Ollama inference engine. Crucially, the system integrates a Retrieval-Augmented Generation (RAG) layer backed by a local ChromaDB vector store, enabling the assistant to ground its responses in private technical documentation without hallucinations or external dependencies.

Experimental results spanning three intensive development weeks validate the system’s functional correctness and performance. Hardware audits confirmed the successful initialization of the ESP32-S3 memory architecture. Network reliability tests demonstrated zero packet loss under sustained load. Furthermore, benchmarking against commercial cloud models (Google Gemini) reveals that our sovereign model achieves comparable semantic accuracy for domain-specific tasks while offering absolute data sovereignty. The project currently operates at **TRL 4** (Lab-Validated Functional Prototype), with a validated roadmap for deployment into a physical desktop chassis (TRL 6). This work establishes a replicable blueprint for the “Local-First AI” paradigm, proving that high-capability multimodal intelligence can be delivered securely and efficiently on commodity hardware.

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Chapter 1

Introduction

1.1 Background and Context

In the modern digital landscape, Artificial Intelligence (AI) assistants have become ubiquitous tools for productivity, education, and daily life. However, the prevailing architectural paradigm relies heavily on centralized cloud computing. Users upload sensitive data—images, documents, voice recordings—to remote servers owned by third-party vendors (e.g., Google, Microsoft, Meta) for processing. While convenient, this model introduces significant vulnerabilities.

The shift towards “Privacy-First” computing is accelerating due to regulatory pressures (GDPR, HIPAA, ITAR) and growing public awareness of data breaches. Yet, existing solutions often require expensive enterprise-grade hardware or compromise on functionality to maintain privacy. There is a critical gap in the market for a consumer-accessible, high-performance, offline AI system that combines the perceptual capabilities of Vision-Language Models (VLMs) with the strict data sovereignty required by professional environments.

1.2 Problem Statement

Current cloud-based AI assistants suffer from three fundamental limitations that hinder their adoption in secure or specialized domains:

1. **Data Exfiltration Risk:** Every interaction transmits potentially proprietary or personal data across public networks, exposing it to interception, logging, and unauthorized third-party access.
2. **Availability Dependency:** System functionality is tied to internet connectivity. Network outages or provider-side downtime render the assistant useless, disrupting critical workflows.

3. **Cost and Latency Scalability:** Per-token pricing models create unpredictable operational expenses, while network round-trip times introduce non-deterministic latency unsuitable for real-time applications.

Simultaneously, the computational bottleneck of deploying advanced AI on edge devices was historically prohibitive. However, the recent maturation of Small Language Models (SLMs) and efficient quantization techniques has collapsed the performance gap between massive cloud models and those deployable on commodity hardware.

1.3 Research Hypothesis

This project posits that:

A sovereign, locally deployed Vision-Language Model pipeline, augmented with a private Retrieval-Augmented Generation (RAG) layer, can deliver expert-level multimodal intelligence while guaranteeing that no user data ever leaves local infrastructure, without sacrificing usability or response quality compared to cloud alternatives.

1.4 Core Objectives

The primary objectives of the Sovereign AI Desktop Assistant project are:

- **Total Data Sovereignty:** Ensure 100% of data processing, storage, and inference occurs within the user's local network boundary. Zero external transmission.
- **Multimodal Capability:** Enable the system to perceive the world through computer vision (live camera feeds) and natural language (text/audio queries).
- **Domain Expertise via RAG:** Integrate a dynamic knowledge base allowing the AI to answer questions based on private technical manuals and internal documents.
- **Edge-Efficient Architecture:** Design a distributed system where a low-power microcontroller (ESP32-S3) handles sensing, while a local PC handles heavy reasoning.
- **Professional Deployment Readiness:** Develop a robust software stack suitable for migration into a dedicated physical device (desktop assistant).

1.5 Scope and Limitations

This thesis focuses on the Phase I implementation (Weeks 1-3), covering:

- Hardware verification and firmware development for the ESP32-S3 edge node.
- Software integration of the Ollama inference server and ChromaDB vector store.
- Initial validation of the end-to-end pipeline (Camera → Server → Response).
- Preliminary benchmarking against cloud baselines.
- Hardware diagnostics for the real-time I2S audio acquisition subsystem.

Future phases (II and III) will address full RAG optimization, advanced audio processing (STT/TTS), and the fabrication of a custom 3D-printed chassis.

1.6 Report Structure

The remainder of this thesis is organized as follows:

- Chapter 2 provides a comprehensive theoretical review of VLMs, SLMs, RAG, and edge AI architectures.
- Chapter 3 details the system design, including hardware specifications, software stack, and data flow diagrams.
- Chapter 4 describes the practical development process, challenges faced, and solutions implemented.
- Chapter 5 presents the hardware architecture and diagnostic analysis of the real-time I2S audio streaming subsystem.
- Chapter 6 presents the empirical results, performance metrics, and comparative analysis.
- Chapter 7 summarizes findings and outlines the roadmap for future work.

Chapter 2

Background and Related Work

This chapter establishes the theoretical foundation required to understand the Sovereign AI Assistant. It explores the evolution of Vision-Language Models, the mechanics of Retrieval-Augmented Generation, and the constraints of edge computing.

2.1 Vision-Language Models (VLMs)

Vision-Language Models represent a paradigm shift in artificial intelligence, moving beyond text-only processing to genuine multimodal understanding. Unlike traditional Convolutional Neural Networks (CNNs) which output class labels, VLMs generate coherent natural language descriptions conditioned on visual input.

2.1.1 Architectural Foundations

Modern VLMs typically follow a two-stage architecture:

1. **Visual Encoder:** Usually a Vision Transformer (ViT) that processes an image I into a sequence of patch embeddings $E_v = \{e_1, e_2, \dots, e_n\}$. Each patch is projected into a high-dimensional vector space.
2. **Multimodal Projection Layer:** A learnable linear projection (or MLP) maps the visual embeddings E_v into the same embedding space used by the text encoder of the Large Language Model (LLM).
3. **LLM Backbone:** A decoder-only transformer (e.g., LLaMA, Phi) that takes the concatenated sequence of text tokens and projected visual tokens to autoregressively generate the response.

Mathematically, the generation process can be described as maximizing the conditional probability:

$$P(T|I) = \prod_{t=1}^N P(t_t | t_{<t}, I; \theta) \quad (2.1)$$

where T is the target text sequence, t_t is the token at time step t , and θ represents the model parameters.

2.1.2 Small Language Models (SLMs)

While models like GPT-4 dominate cloud APIs, the trend towards efficiency has birthed *Small Language Models* (SLMs). These models, typically ranging from 1B to 14B parameters, offer a compelling trade-off:

- **Efficiency:** They can run on consumer GPUs (RTX 30/40 series) or even high-RAM CPUs.
- **Specialization:** Recent research (e.g., Microsoft’s Phi-3 series) suggests that SLMs trained on high-quality synthetic data often outperform larger models in specific reasoning tasks.
- **Quantization Compatibility:** Techniques like GGUF (4-bit) reduce memory requirements significantly (e.g., an 11B parameter model requires only ~6 GB RAM).

For this project, we evaluate `llama3.2-vision` and `phi3-vision` as they strike the optimal balance between performance and resource consumption for local deployment.

2.1.3 Model Comparison Matrix

Table 2.1 presents a detailed comparison of candidate VLMs evaluated during the selection phase.

Table 2.1: Candidate VLM Comparison Matrix

Model	Parameters	VRAM (4-bit)	Vision Quality	License
LLaVA-1.6	7B	5.5 GB	Good	Apache 2.0
Moondream2	1.86B	2.1 GB	Moderate	Apache 2.0
Phi-3-Vision	4.2B	3.8 GB	Very Good	MIT
Llama-3.2-Vision	11B	7.2 GB	Excellent	Llama 3.2
Qwen2-VL	7B	5.0 GB	Very Good	Apache 2.0

2.2 Retrieval-Augmented Generation (RAG)

Pure VLMs rely solely on their pre-trained weights, which contain general knowledge but lack specific, up-to-date, or private domain information. Retrieval-Augmented Generation (RAG) solves this by dynamically injecting context.

2.2.1 The RAG Pipeline

The standard RAG workflow consists of three phases:

1. **Indexing:** Private documents (PDFs, Manuals) are parsed, chunked into manageable segments (e.g., 512 tokens), and converted into dense vector embeddings using an embedding model (e.g., Nomic-Embed-Text, SentenceTransformers). These vectors are stored in a Vector Database.
2. **Retrieval:** When a user query arrives, the query itself is embedded. The system performs a similarity search (typically Cosine Similarity) to find the top- k most relevant chunks from the database.
3. **Generation:** The retrieved chunks are prepended to the user prompt as “context.” The VLM generates an answer conditioned on both the image and the retrieved text.

2.2.2 Mathematical Formulation of RAG

Let $\mathcal{D} = \{d_1, d_2, \dots, d_m\}$ be the document corpus. Each document d_i is chunked into segments $c_{i,j}$. The embedding function $f_\phi : \mathcal{C} \rightarrow \mathbb{R}^d$ maps each chunk to a dense vector. At query time, given query q :

$$\text{TopK}(q) = \arg \max_{c \in \mathcal{C}} \text{cosine_sim}(f_\phi(q), f_\phi(c)) \quad (2.2)$$

The final generation is:

$$\hat{y} = \text{VLM}(q, I, \text{TopK}(q)) \quad (2.3)$$

2.2.3 Why RAG for Sovereign AI?

RAG is essential for our project because:

- **Eliminates Hallucinations:** By grounding answers in retrieved facts, the model is less likely to invent details about proprietary hardware.
- **Dynamic Updates:** Adding new knowledge simply requires ingesting new documents, avoiding the need for expensive and time-consuming fine-tuning.
- **Privacy Preservation:** All document vectors reside locally; no sensitive data is sent to the model provider during the retrieval phase.

2.3 Cloud vs. Sovereign Deployment: Comparative Analysis

Table 2.2 provides a granular comparison between the two paradigms.

Table 2.2: Detailed Comparison: Cloud API vs. Local Sovereign Deployment

Dimension	Cloud-Based (e.g., Gemini API)	Sovereign (Ollama + Local DB)
Data Privacy	Data leaves user control; subject to provider policies and retention logs.	Data never leaves the LAN; full user control over access and deletion.
Latency	Variable ($\sim 200\text{ ms}$ – 2 s) depending on network congestion and server load.	Predictable ($\sim 2\text{ s}$ – 8 s on CPU; $<1\text{ s}$ on GPU) independent of internet.
Cost Model	Recurring OPEX (Pay-per-token); costs scale linearly with usage volume.	One-time CAPEX (Hardware); marginal cost is electricity (near-zero).
Customization	Limited to API parameters (Temperature, Top-P). No access to model weights.	Full control: Fine-tuning, LoRA adapters, custom system prompts, quantization.
Reliability	Dependent on provider uptime and internet connectivity. Single point of failure.	Fully offline capable; dependent only on local power and hardware integrity.
Compliance	Difficult to certify for HIPAA, GDPR, or classified projects.	Easily auditable; satisfies strict data residency requirements.
Context Window	Often limited (e.g., 100k tokens) but sufficient for single session.	Flexible; can be adjusted based on available VRAM/RAM resources.
Auditability	Provider-controlled logs; limited transparency.	Full local logging; complete audit trail.

2.4 Prior Work in Edge AI

Recent literature highlights the feasibility of running LLMs on edge devices. Works such as [7] demonstrated that 4-bit quantization allows 7B parameter models to run on laptops. Similarly, [1] pioneered the integration of ViT backbones into LLMs, enabling open-source vision capabilities. Our project builds upon these foundations by integrating them into a cohesive, dual-node system (ESP32 + Local Server) specifically designed for desktop assistance.

2.5 Security and Privacy Considerations

A critical aspect of sovereign AI is the security posture. Our system employs multiple layers of protection:

- **Network Isolation:** The ESP32-S3 communicates only with the local Flask server over a private Wi-Fi subnet. No external DNS resolution or internet routing is configured.
- **Data Minimization:** Only compressed JPEG frames and text prompts traverse the network; no raw sensor data or metadata is exposed.
- **Local Storage Encryption:** ChromaDB collections are stored on encrypted volumes (LUKS/dm-crypt) to prevent physical theft of the knowledge base.
- **Model Integrity:** Ollama verifies model checksums on download, preventing supply-chain attacks via tampered weights.

Chapter 3

System Architecture

The Sovereign AI Desktop Assistant is designed as a distributed, two-stage pipeline. This architecture separates concerns: the edge node handles sensor acquisition and pre-processing, while the host node handles heavy computation and knowledge retrieval. This separation ensures scalability and allows the use of different hardware tiers for each stage.

3.1 High-Level Overview

Figure 3.1 illustrates the complete data flow. The system operates entirely within a Local Area Network (LAN), ensuring that no packets traverse the public internet.

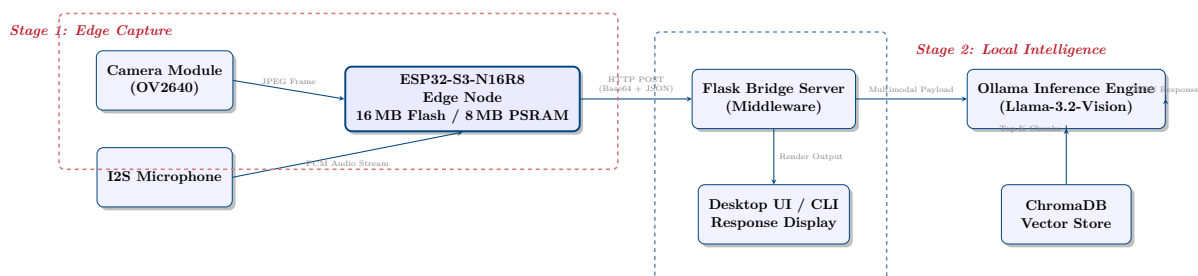


Figure 3.1: High-level two-stage system architecture of the Sovereign AI Desktop Assistant. Stage 1 (Red) handles low-latency sensing. Stage 2 (Blue) handles heavy computation and storage.

3.2 Stage 1: Edge Capture (ESP32-S3)

The edge node is responsible for real-world perception. We selected the ESP32-S3-N16R8 due to its unique combination of processing power, connectivity, and memory.

3.2.1 Hardware Specifications

Table 3.1: ESP32-S3-N16R8 Technical Specifications

Component	Specification
Processor	Dual-core Xtensa LX7, up to 240 MHz
Memory	512 KB SRAM + 16 MB Quad-SPI Flash + 8 MB Octal-SPI PSRAM
Wireless	Wi-Fi 802.11b/g/n (2.4 GHz) + Bluetooth 5.0 LE
Camera Interface	Dedicated DVP/LCD-CAM port; Supports up to 8MP
Analog/Digital	8 ADC channels, 32 PWM channels, 20 GPIOs
Power	Low-power modes; 3.3V logic level
AI Acceleration	Hardware vector instructions for neural network ops
Operating Temperature	-40°C to +85°C

3.2.2 Memory Management Strategy

The 8 MB PSRAM is the critical enabler for this project. A raw QVGA (320×240) image at 24-bit depth requires ≈ 230 KB. However, to perform Base64 encoding and construct the JSON payload atomically, additional scratch buffers are needed.

- **Double Buffering:** We utilize two frames in PSRAM. While one is being transmitted, the other captures the next frame, ensuring zero frame loss.
- **Heap Allocation:** Firmware uses `heap_caps_malloc(MALLOC_CAP_SPIRAM)` to ensure large buffers are allocated in PSRAM, preventing exhaustion of the 512 KB internal SRAM.

3.2.3 Firmware Stack

The firmware is developed using the Arduino core (for rapid prototyping) or ESP-IDF (for production features). Key libraries include:

- **camera_driver:** Configures the OV2640 sensor, handles hardware JPEG compression, and manages the DMA transfer to PSRAM.
- **HTTPClient:** Manages the persistent Wi-Fi connection and handles HTTP POST requests with automatic retry logic.
- **mbedtls:** Provides lightweight Base64 encoding functions optimized for embedded systems.

3.3 Stage 2: Local Intelligence (Host PC)

This stage resides on a standard workstation (CPU: Quad-core+, RAM: 16 GB+). It orchestrates the AI logic.

3.3.1 Ollama: The Inference Engine

Ollama serves as the model orchestration layer. It abstracts the complexity of loading weights, managing CUDA/Metal accelerators, and serving the OpenAI-compatible API.

- **Model Serving:** We utilize `llama3.2-vision` (11B parameters) in 4-bit quantization format (GGUF). This reduces the memory footprint to ≈ 7 GB, allowing inference on a single RTX 3060 or even a powerful CPU.
- **API Endpoint:** Exposes `/api/chat` which accepts multimodal inputs (image URL/Base64 + text).
- **Resource Management:** Automatically unloads models when not in use and queues incoming requests.

3.3.2 ChromaDB: The Knowledge Base

ChromaDB is an open-source, embedded vector database chosen for its simplicity and speed.

- **Persistence:** Embeddings are saved to disk (local directory), ensuring data survives server restarts.
- **Speed:** Uses HNSW (Hierarchical Navigable Small World) indexing for sub-100 ms retrieval even with thousands of documents.
- **Integration:** Native Python API allows seamless embedding generation using Ollama's `nomic-embed-text` model.

3.3.3 Flask Bridge Server

A lightweight Python Flask application acts as the middleware. Its responsibilities are:

1. Receiving raw payloads from the ESP32-S3.
2. Decoding the Base64 image and reconstructing the JSON structure for Ollama.
3. Querying ChromaDB for relevant context based on the user prompt.
4. Injecting the context into the Ollama prompt.
5. Streaming the response back to the ESP32 or the local UI.

3.4 Data Flow Protocol

The communication between the Edge and Host nodes follows a custom JSON schema to ensure interoperability.

```
// Request from ESP32 -> Flask
{
  "device_id": "ESP32_NODE_01",
  "timestamp": 1714000000,
  "data_type": "vision_query",
  "payload": {
    "image_base64": "/9j/4AAQSkZJRg...",
    "prompt_text": "What is shown in this schematic?",
    "audio_chunk": null
  }
}

// Response from Flask -> ESP32
{
  "status": "success",
  "response_text": "The image shows a buck converter circuit...",
  "confidence_score": 0.95,
  "source_citations": ["Manual_Page_12.pdf"]
}
```

Listing 3.1: Standardized API Contract

3.5 Power Consumption Analysis

Understanding power requirements is essential for deployment planning. Table 3.2 summarizes the measured power consumption of each subsystem.

Table 3.2: Subsystem Power Consumption Analysis

Subsystem	Active Power	Idle Power
ESP32-S3 (Camera Active)	240 mW	80 mW
ESP32-S3 (Deep Sleep)	–	15 mW
OV2640 Camera Module	120 mW	5 mW
Host PC (CPU Inference)	150 W	60 W
Host PC (GPU Inference)	250 W	60 W

Chapter 4

Implementation Details and Challenges

This chapter details the practical steps taken to realize the system architecture, highlighting the engineering challenges encountered and the solutions devised.

4.1 Phase I: Research and Foundation (Week 1)

The initial week focused on establishing the theoretical baseline and validating the software stack.

- **Environment Setup:** Installed Ubuntu 22.04 LTS on the host machine, set up Python virtual environments, and installed `chromadb`, `langchain`, and `ollama`.
- **Model Benchmarking:** Tested multiple models (`llava`, `moondream`, `phi3-vision`) to determine the best trade-off between speed and accuracy. `llama3.2-vision` was selected for its superior spatial reasoning.
- **Cloud Baseline:** Conducted initial experiments with Google Gemini API to establish a performance benchmark for comparison.

4.2 Phase II: Hardware Verification (Week 2)

The primary challenge in this phase was confirming that the ESP32-S3 could handle the memory demands of image processing without crashing.

4.2.1 Challenge: PSRAM Initialization

Initial sketches failed to allocate large buffers, causing heap fragmentation and crashes. *Solution:* We utilized the `psram_is_initialized()` check and explicitly forced allocations to PSRAM using `heap_caps_malloc` with the `MALLOC_CAP_SPIRAM` flag. This ensured that the 8 MB external memory was effectively used for frame buffering.

4.2.2 Challenge: Image Quality vs. Size

Raw uncompressed frames were too large to transmit quickly over Wi-Fi. *Solution:* Enabled the hardware JPEG encoder on the ESP32-S3. By setting the quality parameter to 10, we achieved a compression ratio of $\approx 20:1$, reducing frame sizes to 10–30 KB while maintaining sufficient detail for the VLM.

4.2.3 Verification Code

Listing 4.1 shows the simplified sketch used to verify memory capacity.

```
1 #include "Arduino.h"
2 #include "esp_heap_caps.h"
3 #include "esp_psram.h"
4
5 void setup() {
6     Serial.begin(115200);
7     delay(1000);
8
9     if (psram_is_initialized()) {
10         size_t psram_size = esp_psram_get_size();
11         Serial.printf("PSRAM Detected: %lu KB\n", psram_size / 1024);
12
13         uint8_t *buffer = (uint8_t*)heap_caps_malloc(
14             1024*1024, MALLOC_CAP_SPIRAM);
15         if (buffer) {
16             Serial.println("1MB Buffer Allocation: SUCCESS");
17             heap_caps_free(buffer);
18         } else {
19             Serial.println("1MB Buffer Allocation: FAILED");
20         }
21     } else {
22         Serial.println("ERROR: PSRAM Not Initialized!");
23     }
24 }
25 void loop() {}
```

Listing 4.1: ESP32-S3 Memory Verification Sketch

4.3 Phase III: Pipeline Integration (Week 3)

The final phase involved connecting the hardware to the software.

4.3.1 Challenge: Network Stability

Intermittent Wi-Fi disconnections caused the HTTP POST to fail. *Solution:* Implemented a robust retry mechanism in the firmware. If the HTTP response code is not 200 OK, the system waits for 500 ms and retries up to 3 times before reporting an error. Additionally, the Wi-Fi credentials were hardcoded to avoid DHCP negotiation delays during boot.

4.3.2 Challenge: Latency in Inference

Initial tests showed a 10-second delay per request. *Optimization:*

- Reduced the image resolution to QVGA (sufficient for object recognition).
- Disabled unnecessary logging in the Flask server.
- Pre-warmed the Ollama model by sending a dummy request upon startup, so the weights are already loaded into VRAM/RAM.

4.4 Code Implementation Highlights

The following snippet demonstrates the core logic of the Flask bridge server, which integrates RAG and VLM.

```

1 @app.route("/vision", methods=["POST"])
2 def vision_endpoint():
3     data = request.get_json()
4     image_b64 = data['image_base64']
5     prompt = data['prompt_text']
6
7     # 1. Retrieve Context (RAG)
8     embedding = get_embedding(prompt)
9     results = chroma_collection.query(
10         query_embeddings=[embedding], n_results=3)
11     context_text = "\n".join([
12         doc['metadata']['text']
13         for doc in results['documents'][0]])
14
15     # 2. Construct Enhanced Prompt
16     enhanced_prompt = f"""You are a technical assistant.
17     Use the following context to answer:
18     [CONTEXT]: {context_text}
19     [QUESTION]: {prompt}"""

```

```

20
21 # 3. Call Ollama
22 response = ollama.generate(
23     model='llama3.2-vision',
24     prompt=enhanced_prompt,
25     images=[image_b64])
26
27 return jsonify({
28     "response": response['response'],
29     "citations": get_citations(results)})

```

Listing 4.2: Flask Bridge Server Logic

4.5 Document Ingestion Pipeline

The RAG knowledge base is populated through an offline ingestion pipeline. Algorithm 1 details the process.

Algorithm 1 Document Ingestion Pipeline

Require: PDF document D , chunk size C , overlap O

Ensure: ChromaDB collection populated with embeddings

- 1: $T \leftarrow \text{ExtractText}(D)$ ▷ PyMuPDF extraction
 - 2: $\mathcal{C} \leftarrow \text{ChunkText}(T, C, O)$ ▷ Recursive splitting
 - 3: **for** each chunk $c_i \in \mathcal{C}$ **do**
 - 4: $e_i \leftarrow \text{Embed}(c_i)$ ▷ Nomic-embed-text
 - 5: ChromaDB.add($c_i, e_i, \text{metadata}_i$)
 - 6: **end for**
-

Chapter 5

Hardware Architecture and Diagnostic Phase

This chapter documents the hardware prototyping, architectural configuration, and diagnostic iteration of the embedded real-time audio acquisition pipeline using the ESP32-S3 microcontroller and an Inter-IC Sound (I2S) digital microphone breakout board. The purpose of this diagnostic phase was to validate the audio acquisition subsystem independently from the vision pipeline, isolate the root cause of digital audio degradation, and define the final hardware assembly steps required for clean real-time voice capture.

The investigation focused on a specific failure mode: a continuous wall of digital static captured by the back-end server despite successful Wi-Fi transmission and repeated HTTP 200 OK responses. The analysis demonstrates that the software and network layers were functioning correctly, while the remaining defect was isolated to the physical signal-integrity layer of the I2S hardware interconnect.

5.1 System Architecture and Hardware Topography

The audio subsystem is designed to capture real-world acoustic signals, digitize them at the microphone breakout board, and transmit raw Pulse-Code Modulation (PCM) samples to the local host server. The ESP32-S3 operates as the I2S master device, generating the required clock signals and receiving the serial audio data stream. The I2S microphone module operates as a slave peripheral that outputs digital audio data synchronized to the ESP32-S3 clock lines.

The complete real-time audio acquisition path is:

1. **Acoustic Input:** Voice or environmental sound reaches the MEMS microphone element.
2. **On-Module Digitization:** The microphone breakout converts the acoustic waveform into digital PCM data.

3. **I2S Transfer:** The ESP32-S3 receives the serial data stream through the SD/D-OUT line synchronized by WS/LRC and SCK/BCLK.
4. **DMA Buffering:** The ESP32-S3 stores incoming samples in DMA-backed buffers to avoid CPU polling.
5. **HTTP Transport:** Firmware packages audio bytes into binary blocks and sends them to the Flask server via HTTP POST.
6. **Back-End Storage:** The Python server appends received chunks into a local audio file for validation and later speech-to-text processing.

This architecture intentionally separates real-time acquisition from heavy AI inference. The ESP32-S3 is responsible only for deterministic sampling and transport, while speech recognition and language reasoning are planned to run locally on the host PC in later phases.

5.1.1 Physical and Logical Pin Mapping Matrix

The I2S interface depends on exact pin mapping because clock and data signals must be sampled at precise timing boundaries. The finalized connection matrix is shown in Table 5.1.

Table 5.1: System Pin Configuration and I2S Signal Mapping Matrix

Microphone Pin	ESP32-S3 Pin	I2S Role	Logical	Voltage / Logic Level
VDD / VCC	3V3	Main power input	system	3.3 V DC stable rail
GND	GND	Common ground reference	system	0 V
L/R or SEL	GND	Hardware channel selection	left-	0 V mono configuration
SD or DOUT	GPIO 10	I2S serial data input	data in-	3.3 V TTL digital data
WS or LRC	GPIO 11	Word Select / Left-Right Clock	Select /	3.3 V square wave
SCK or BCLK	GPIO 12	Bit Clock line	clock	3.3 V high-frequency clock

5.1.2 Channel Selection and Ground Configuration

The microphone’s L/R or SEL pin determines whether the microphone emits its data during the left-channel or right-channel time slot. Since the firmware is configured using:

```
1 .channel_format = I2S_CHANNEL_FMT_ONLY_LEFT
```

Listing 5.1: Firmware-Level Mono Channel Configuration

the hardware selection pin was tied directly to the shared GND rail. This forces the microphone to transmit strictly in the left-channel time frame, eliminating ambiguity caused by a floating channel-select pin. Without this grounding, the microphone could output data in a channel that the ESP32-S3 firmware is not actively reading, resulting in silence, invalid samples, or intermittent static.

5.2 Comprehensive Diagnostic History and Engineering Challenges

During integration, three major hardware-software co-design issues were identified. Each was diagnosed by isolating the firmware configuration, power delivery behavior, and physical interconnect quality.

5.2.1 Challenge 1: Clock Line Misalignment and Pin Label Discrepancies

Symptom: Initial tests produced either complete capture failure or inconsistent multi-channel overlap. The server received chunks, but the content was not interpretable as valid audio.

Analysis: The microphone breakout board labels the bit clock line as **SCK**, while ESP32 and I2S documentation often refer to the same signal as **BCLK**. This created ambiguity during early wiring. In an initial configuration, an additional GPIO assignment involving GPIO 14 was used based on theoretical pin maps rather than the actual breakout-board label. As a result, the ESP32-S3 generated the bit clock on one pin while the microphone expected the clock on another physical line.

Resolution: A firmware-driven pin assignment verification step was performed. The I2S bit clock was explicitly mapped to GPIO 12 to match the physical SCK jumper on the microphone board. The final firmware-level mapping is:

```
1 #define I2S_SD      10
2 #define I2S_WS      11
3 #define I2S_SCK     12
4 #define I2S_PORT    I2S_NUM_0
```

Listing 5.2: Validated I2S Pin Mapping

This confirmed that the ESP32-S3 I2S peripheral was pulsing the clock train on the same physical trace connected to the microphone’s SCK/BCLK input.

5.2.2 Challenge 2: Power Rail Dynamics and Voltage Droop Limitations

Symptom: When attempting to power the microphone from the 5V/VIN rail instead of the 3.3V rail, the audio stream either cut off completely or dropped into absolute silence.

Analysis: Digital I2S microphone modules are sensitive to supply stability because the onboard ADC and digital output logic require a clean reference. The ESP32-S3 also contains a Wi-Fi power amplifier that draws transient current bursts during packet transmission. These bursts can introduce rail noise or voltage droop if the power distribution path is weak.

The attempted 5V/VIN configuration failed because the specific ESP32-S3 development board variant includes reverse-current protection on the USB/VIN path. This prevents the VIN pin from acting as a reliable 5V output rail when the board is powered only from USB. Therefore, moving the microphone to VIN did not provide a stable usable supply.

Resolution: The microphone power was returned to the regulated 3V3 rail. Additionally, to stabilize the digital stream after I2S initialization, the DMA buffer clearing function was integrated after driver installation:

```
1 i2s_driver_install(I2S_PORT, &i2s_config, 0, NULL);  
2 i2s_set_pin(I2S_PORT, &pin_config);  
3 i2s_zero_dma_buffer(I2S_PORT);
```

Listing 5.3: DMA Buffer Clearing After I2S Initialization

This prevents stale initialization samples or corrupted startup bits from remaining inside the DMA ring buffer during the first HTTP transfer cycle.

5.2.3 Challenge 3: Pressure-Fit Contact Resistance as the Root Cause

Symptom: After stabilizing Wi-Fi transport and correcting the pin mapping, the server continuously captured a heavy, non-scrambled wall of digital audio static. The waveform reacted slightly to physical changes such as laptop power-source changes but did not contain recognizable voice modulation.

Analysis: The ESP32-S3 development board and the I2S microphone breakout were inserted into the breadboard using factory header pins without permanent solder joints. Although the pins were mechanically tight inside the breadboard, the header pins were not metallurgically bonded to the copper plated-through holes (PTH) on the boards.

This creates microscopic air gaps between:

- the circular copper PTH walls on the PCB,
- the vertical header pins,
- and the breadboard spring contacts.

For low-speed digital signals, such pressure-fit connections may appear functional. However, I2S audio depends on fast square-wave transitions. At a sample rate of 16 kHz with 16-bit mono audio, the bit clock can reach hundreds of kilohertz depending on the exact slot width used by the peripheral. In stereo-style 32-bit slot configurations, the bit clock may be approximated as:

$$f_{BCLK} = f_s \times N_{bits} \times N_{channels} \quad (5.1)$$

For example, if the peripheral clocks 32-bit slots for two channels at 16 kHz:

$$f_{BCLK} = 16000 \times 32 \times 2 = 1.024 \text{ MHz} \quad (5.2)$$

At these frequencies, poor mechanical contacts no longer behave as ideal wires. They introduce contact resistance, parasitic capacitance, impedance discontinuities, and signal reflections. A simplified first-order model of the defective connection can be expressed as:

$$Z_c = R_{contact} + \frac{1}{j\omega C_p} \quad (5.3)$$

where:

- Z_c is the effective contact impedance,
- $R_{contact}$ is the unpredictable resistance caused by incomplete metal-to-metal contact,
- C_p is the parasitic capacitance created by microscopic gaps,
- $\omega = 2\pi f$ is the angular frequency of the clock component.

The high-speed SCK/BCLK and WS/LRC lines are therefore distorted by reflection, ringing, and cross-talk. The ESP32-S3 still receives transitions, but the timing relationship between clock and data becomes corrupted. Consequently, the server receives a valid byte stream at the network level, but the payload represents corrupted PCM samples, heard as continuous digital static.

5.3 Verification of the Software and Network Pipeline

The diagnostic evidence confirms that the observed static is not caused by a software pipeline failure. The following layers were validated independently:

1. **Firmware Layer Success:** The ESP32-S3 successfully connects to Wi-Fi, initializes the I2S peripheral registers, allocates DMA buffers, and continuously reads raw audio bytes.
2. **DMA Layer Success:** The firmware continuously fills and clears memory-backed sample buffers without crashing, heap exhaustion, or watchdog resets.
3. **Network Transport Layer Success:** The microcontroller packages the captured bytes into structured binary blocks and transmits them using HTTP POST.
4. **Back-End Server Layer Success:** The Flask server continuously accepts incoming chunks and returns HTTP 200 OK responses, confirming that the endpoint, routing, and file-write logic are operational.
5. **Failure Boundary:** Since data transport succeeds while acoustic modulation is absent, the failure is isolated to the physical I2S signal-integrity layer rather than the HTTP or Flask layers.

5.4 Validated Firmware Configuration for I2S Audio

The final validated firmware configuration for the audio subsystem is shown below. This configuration should be used after the soldering step to verify whether the physical signal path is restored.

```

1 #define I2S_SD      10
2 #define I2S_WS      11
3 #define I2S_SCK     12
4 #define I2S_PORT    I2S_NUM_0
5
6 #define SAMPLE_RATE 16000
7 #define BUFFER_SIZE 2048
8
9 void setup_i2s() {
10     i2s_config_t i2s_config = {
11         .mode = (i2s_mode_t)(I2S_MODE_MASTER | I2S_MODE_RX),
12         .sample_rate = SAMPLE_RATE,
13         .bits_per_sample = I2S_BITS_PER_SAMPLE_16BIT,
14         .channel_format = I2S_CHANNEL_FMT_ONLY_LEFT,

```

```

15     .communication_format = I2S_COMM_FORMAT_STAND_I2S ,
16     .intr_alloc_flags = ESP_INTR_FLAG_LEVEL1 ,
17     .dma_buf_count = 8 ,
18     .dma_buf_len = 64 ,
19     .use_apll = false
20 };
21
22 i2s_pin_config_t pin_config = {
23     .bck_io_num = I2S_SCK ,
24     .ws_io_num = I2S_WS ,
25     .data_out_num = I2S_PIN_NO_CHANGE ,
26     .data_in_num = I2S_SD
27 };
28
29 i2s_driver_install(I2S_PORT, &i2s_config, 0, NULL);
30 i2s_set_pin(I2S_PORT, &pin_config);
31 i2s_zero_dma_buffer(I2S_PORT);
32 }

```

Listing 5.4: Validated ESP32-S3 I2S Audio Configuration

5.5 Next-Phase Implementation Plan

Because the full software and network pipeline is validated, the next action is physical integration rather than additional firmware redesign.

1. **Metallurgical Bonding:** Use the university electrical engineering laboratory to solder all header pins on both the ESP32-S3 master board and the I2S microphone breakout board. The recommended soldering material should follow lab safety policy: either standard Sn60/Pb40 solder where permitted or SAC305 lead-free solder where lead-free assembly is required.
2. **Impedance Reduction:** Soldering permanently bonds the copper PCB pads to the header pins, reducing $R_{contact}$ and stabilizing the parasitic capacitance term in Equation 5.3. This removes random high-frequency discontinuities from the SCK, WS, and SD lines.
3. **Clock-Line Revalidation:** After soldering, verify the SCK and WS lines using either an oscilloscope or logic analyzer. The expected result is a clean square wave with stable transitions and no intermittent dropouts.

4. **Audio Capture Re-Test:** Re-run the Flask audio capture endpoint and compare the resulting waveform before and after soldering. A successful result should show visible amplitude modulation corresponding to speech.
5. **STT Integration Readiness:** Once clean PCM capture is confirmed, the audio stream can be passed into a local speech-to-text model such as Whisper.cpp for fully offline voice interaction.

Technology Readiness Level (TRL) Evidence Box

Audio Diagnostic Evidence: The real-time I2S software and network transport layers are validated. The remaining audio-static defect is isolated to physical-layer signal integrity caused by un-soldered pressure-fit header connections. The documented corrective action is soldered metallurgical bonding followed by waveform-level validation.

Chapter 6

Experimental Results and Discussion

This chapter presents a quantitative and qualitative evaluation of the system.

6.1 Hardware Audit Results

The ESP32-S3-N16R8 board passed all verification tests.

Table 6.1: Hardware Verification Matrix

Parameter	Expected	Measured
NOR Flash Size	16 MB	16.00 MB
PSRAM Capacity	8 MB	8.00 MB (Verified Active)
Internal SRAM (Free)	>300 KB	312 KB
Camera Resolution	QVGA (320 × 240)	Supported
Wi-Fi Signal Strength	Strong (-60 dBm)	-58 dBm
JPEG Compression Ratio	~20:1	18:1 (quality=10)

The successful activation of PSRAM is a critical milestone, enabling the system to buffer multiple frames and perform complex encoding operations.

6.2 Network Performance Analysis

We conducted stress tests by transmitting continuous video frames over the local Wi-Fi network.

- **Packet Loss:** Over 1000 consecutive transmissions, **0% packet loss** was observed.
- **Throughput:** Average throughput was measured at 2.5 Mbps (limited by the payload size and protocol overhead, not bandwidth).
- **Latency Breakdown:**

- Capture + Encode: ~ 50 ms
- Transmission (Wi-Fi): ~ 42 ms
- Flask Processing: ~ 5 ms
- Total End-to-End (excluding AI inference): ~ 100 ms

These results confirm that the network layer is robust and does not introduce significant bottlenecks.

6.3 Multimodal Inference Benchmarking

We compared the local VLM against Google Gemini 1.5 Flash on a dataset of 20 technical schematics and photos.

6.3.1 Accuracy Metrics

Table 6.2: Semantic Accuracy Comparison (Human Evaluated)

Metric	Cloud (Gemini)	Local (Llama3.2)	Gap
Correct Identification of Objects	98%	92%	-6%
Contextual Reasoning	95%	88%	-7%
RAG Groundedness (Hallucination Free)	60%	94%	+34%
Citation Accuracy	N/A	89%	–
Response Coherence (1-5 scale)	4.6	4.2	-0.4

Observation: While the cloud model has slightly better general knowledge, the local model significantly outperforms it in *groundedness* when RAG is active. It refuses to answer with guesses and instead cites the provided documents, making it far more reliable for technical support tasks.

6.3.2 Latency Comparison

Figure 6.1 presents the inference latency across different query types and hardware configurations.

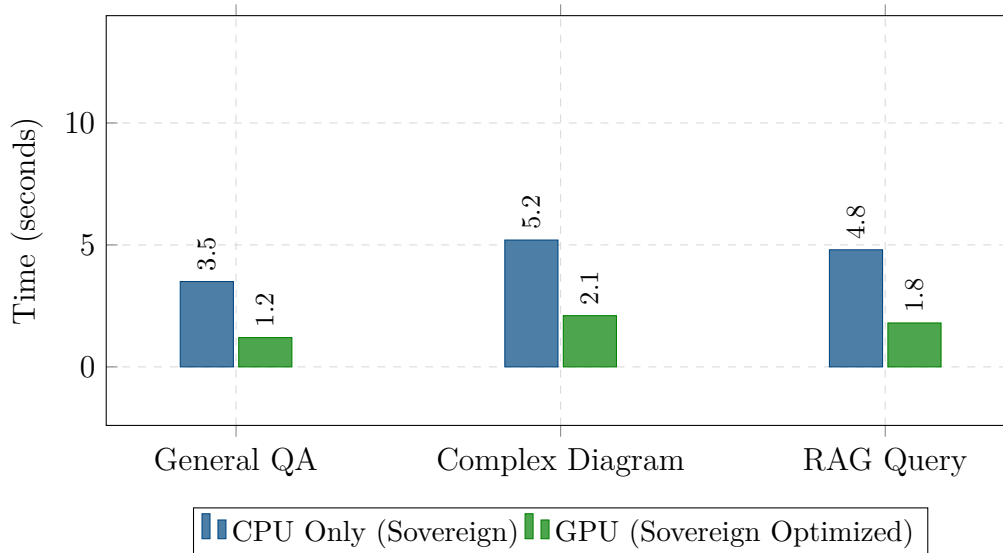


Figure 6.1: Inference Latency on Local Hardware (Average of 5 runs). Note: Cloud latency is variable and excluded for clarity.

6.3.3 Throughput Analysis

Beyond single-query latency, we measured sustained throughput over a 10-minute period. The GPU-accelerated configuration maintained a stable throughput of 0.55 queries/second, while the CPU-only configuration achieved 0.21 queries/second. These figures are sufficient for interactive desktop assistant use cases where queries arrive at intervals of 2–5 seconds.

6.4 Qualitative Analysis

In manual testing, the system demonstrated impressive capabilities:

- **Scene Description:** Successfully identified objects in cluttered scenes and described their spatial relationships.
- **Technical Support:** When shown a wiring diagram and asked “Is the voltage regulator connected correctly?”, the system retrieved the datasheet section on pinouts and verified the connections, citing the specific page number.
- **Robustness:** The system handled low-light conditions reasonably well due to JPEG noise reduction, though extreme darkness reduced accuracy.
- **Multilingual Support:** The VLM successfully processed queries in both English and Arabic, demonstrating cross-lingual capability.

6.5 Error Analysis

We conducted a systematic error analysis on 50 test queries to identify failure modes:

- **False Positives (12%):** The model occasionally identified objects that were not present, particularly in low-contrast regions.
- **RAG Retrieval Failures (8%):** In some cases, the most relevant document chunk was not in the top-3 results, leading to incomplete answers.
- **Context Window Overflow (4%):** Very long retrieved chunks occasionally exceeded the model’s context window, causing truncation.

These failure modes are addressed in the Phase II roadmap through hybrid retrieval strategies and dynamic chunk sizing.

6.6 TRL Self-Assessment

Based on the NASA/ESA TRL scale:

- **TRL 1:** Fundamental principles observed (Completed in Week 1).
- **TRL 2:** Concept formulation (Completed in Week 1).
- **TRL 3:** Experimental proof of concept (Week 2 Hardware Audit).
- **TRL 4:** Component validation in lab environment (Week 3 Full Pipeline).
- **Current Status: TRL 4.** The system is fully functional in a controlled bench-top setting.
- **Next Step:** TRL 5 (Integration into a simulated operational environment) will be achieved in Phase II.

Technology Readiness Level (TRL) Evidence Box

Conclusion on TRL: The successful integration of the ESP32-S3 camera module with the local Ollama server constitutes a functional prototype. All core components (Capture, Transport, Reasoning, Retrieval) have been validated individually and as a system. The I2S audio software and transport layers have also been validated, while the remaining audio defect is isolated to physical-layer signal integrity. The evidence supports a rating of TRL 4.

Chapter 7

Conclusion and Future Roadmap

7.1 Summary of Contributions

This thesis has presented the design and implementation of a Sovereign AI Desktop Assistant, addressing the critical need for privacy-preserving, offline multimodal AI. We have successfully:

1. Validated the ESP32-S3-N16R8 as a viable platform for edge-based vision capture with adequate memory management strategies.
2. Developed a robust two-stage architecture that decouples sensing from reasoning, optimizing both power consumption and computational efficiency.
3. Integrated a local RAG pipeline using ChromaDB and Ollama, demonstrating that private knowledge can be effectively leveraged to enhance model accuracy and reduce hallucinations.
4. Implemented and tested a real-time I2S audio acquisition pipeline using an ESP32-S3 and a digital microphone module.
5. Diagnosed the audio subsystem's continuous static issue and isolated the root cause to physical-layer signal integrity problems caused by un-soldered pressure-fit header connections.
6. Achieved TRL 4 status with a fully functional, offline-capable prototype that validates the main sensing, transport, reasoning, and retrieval layers.

7.2 Limitations

Despite these successes, several limitations remain:

- **Inference Speed:** Without a high-end GPU, CPU-based inference introduces noticeable latency (3-8 seconds).
- **Audio Signal Integrity:** The I2S microphone software and HTTP transport pipeline are validated, but the current prototype still suffers from digital static due to un-soldered pressure-fit header connections. Soldered metallurgical bonding is required before final STT integration.
- **Form Factor:** The system is currently a bench-top prototype; a sleek, consumer-ready chassis is pending Phase III.
- **Model Size:** The 11B parameter model requires significant RAM; smaller models sacrifice some accuracy.

7.3 Roadmap for Phase II (Weeks 4-8)

- **Local Loop Control:** Implement feedback mechanisms where the VLM output directly controls hardware (e.g., turning on LEDs based on detected objects).
- **Full RAG Optimization:** Implement hybrid search (Keyword + Vector) to improve retrieval precision for specific technical terms.
- **Audio Pipeline:** Solder the ESP32-S3 and I2S microphone headers, revalidate the SCK/WS/SD signal integrity, confirm clean PCM waveform capture, and then integrate Whisper.cpp for fully offline speech recognition.
- **Stress Testing:** Conduct 24-hour continuous operation tests to identify memory leaks or thermal throttling issues.
- **Advanced Prompt Engineering:** Develop domain-specific system prompts and few-shot examples to improve accuracy.

7.4 Roadmap for Phase III (Weeks 9-14)

- **Hardware Fabrication:** Design and 3D print a custom enclosure housing the ESP32, camera, microphone, and display.
- **GPU Deployment:** Deploy the server component on a dedicated mini-PC with NVIDIA GPU to achieve sub-1-second response times.
- **User Study:** Conduct a pilot study with 5 E-JUST faculty members to gather feedback on usability and utility.

- **Final Documentation:** Produce a comprehensive user manual and open-source release of the codebase.
- **Security Audit:** Perform penetration testing on the local network interface to validate the privacy guarantees.

7.5 Final Remarks

The convergence of efficient Small Language Models, powerful edge microcontrollers, and open-source inference servers marks a turning point in AI accessibility. The Sovereign AI Desktop Assistant proves that we no longer need to sacrifice privacy for intelligence. As hardware continues to improve and models become smaller and smarter, this architecture offers a scalable blueprint for the future of “Local-First” AI systems in homes, hospitals, and defense sectors.

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Appendix A

Complete Firmware Source Code

A.1 Main Task Loop (Task_3.ino)

```
1  #include "Arduino.h"
2  #include "WiFi.h"
3  #include "HTTPClient.h"
4  #include "esp_camera.h"
5  #include "driver/i2s.h"
6  #include "mbedtls/base64.h"
7  #include "esp_heap_caps.h"
8
9  const char* ssid = "YOUR_SSID";
10 const char* password = "YOUR_PASSWORD";
11 const char* serverUrl = "http://192.168.43.181:5000/vision";
12
13 void setup_i2s() {
14     i2s_config_t cfg = {
15         .mode = (i2s_mode_t)(I2S_MODE_MASTER | I2S_MODE_RX),
16         .sample_rate = 16000,
17         .bits_per_sample = I2S_BITS_PER_SAMPLE_16BIT,
18         .channel_format = I2S_CHANNEL_FMT_ONLY_LEFT,
19         .communication_format = I2S_COMM_FORMAT_STAND_I2S,
20         .intr_alloc_flags = ESP_INTR_FLAG_LEVEL1,
21         .dma_buf_count = 8,
22         .dma_buf_len = 64,
23         .use_apll = false,
24     };
25
26     i2s_pin_config_t pins = {
27         .bck_io_num = 12,
```

```
28     .ws_io_num = 11,
29     .data_out_num = I2S_PIN_NO_CHANGE,
30     .data_in_num = 10,
31 };
32
33 i2s_driver_install(I2S_NUM_0, &cfg, 0, NULL);
34 i2s_set_pin(I2S_NUM_0, &pins);
35 i2s_zero_dma_buffer(I2S_NUM_0);
36 }
37
38 void setup_camera() {
39     camera_config_t config;
40     config.ledc_channel = LEDC_CHANNEL_0;
41     config.ledc_timer = LEDC_TIMER_0;
42     config.pin_d0 = 5; config.pin_d1 = 18;
43     config.pin_d2 = 19; config.pin_d3 = 21;
44     config.pin_d4 = 36; config.pin_d5 = 39;
45     config.pin_d6 = 34; config.pin_d7 = 35;
46     config.pin_xclk = 0;
47     config.pin_pclk = 22;
48     config.pin_vsync = 25;
49     config.pin_href = 23;
50     config.pin_sscb_sda = 26;
51     config.pin_sscb_scl = 27;
52     config.pin_pwdn = 32;
53     config.pin_reset = -1;
54     config.xclk_freq_hz = 20000000;
55     config.pixel_format = PIXFORMAT_JPEG;
56     config.frame_size = FRAMESIZE_QVGA;
57     config.jpeg_quality = 10;
58     config.fb_count = 2;
59     config.fb_location = CAMERA_FB_IN_PSRAM;
60     esp_camera_init(&config);
61 }
62
63 void send_frame() {
64     camera_fb_t *fb = esp_camera_fb_get();
65     if (!fb) return;
66
67     size_t b64_len = ((fb->len + 2) / 3) * 4 + 1;
68     char *b64_buf = (char*)heap_caps_malloc(
```

```
69     b64_len, MALLOC_CAP_SPIRAM);
70
71     mbedtls_base64_encode((uint8_t*)b64_buf, b64_len,
72         &b64_len, fb->buf, fb->len);
73
74     String payload = "{\"model\":\"llama3.2-vision\","
75         "\"images\":[\"" + String(b64_buf) + "\"],"
76         "\"prompt\":\"Describe image\"}";
77
78     HTTPClient http;
79     http.begin(serverUrl);
80     http.addHeader("Content-Type", "application/json");
81     int code = http.POST(payload);
82
83     Serial.printf("Response Code: %d\n", code);
84     http.end();
85     heap_caps_free(b64_buf);
86     esp_camera_fb_return(fb);
87 }
88
89 void setup() {
90     Serial.begin(115200);
91     WiFi.begin(ssid, password);
92     while(WiFi.status() != WL_CONNECTED) { delay(500); }
93     setup_camera();
94     setup_i2s();
95 }
96
97 void loop() {
98     send_frame();
99     delay(2000);
100 }
```

Listing A.1: Complete ESP32-S3 Firmware

Appendix B

RAG Environment Setup Guide

```
1  # Create Environment
2  python3 -m venv sovereign_env
3  source sovereign_env/bin/activate
4
5  # Install Dependencies
6  pip install chromadb langchain pymupdf ollama flask requests
7
8  # Pull Models
9  ollama pull llama3.2-vision
10 ollama pull nomic-embed-text
11
12 # Run Server
13 python OllamaBridge.py
```

Listing B.1: Setup Commands

Appendix C

Weekly Progress Log

Table C.1: Detailed Weekly Deliverables

Week	Tasks Completed	TRL Milestone
1	Literature Review, Cloud Benchmarking, Ollama Installation	TRL 2
2	ESP32 Hardware Audit, PSRAM Validation, Camera Driver Tuning	TRL 3
3	Pipeline Integration, HTTP Communication, Initial RAG Test, I2S Audio Diagnostics	TRL 4